

InfoBento Product Brief

See what matters. Skip the spiral.

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A solar-powered B&W eInk display that sits on your counter, desk, or shelf. Weather, a countdown, a quote — visible at a glance from across the room, in crisp black ink on paper-white. No cables, no account, no batteries to swap, no buttons to press.

Product Overview

InfoBento is a small, calm surface for the room. The information you check most often — weather, the next thing on your calendar, days until something you're looking forward to — sits there in sharp B&W eInk, visible at a glance. Configure once on a web page; it sips light from the window and refreshes on its own.

Target price: \$30-40 via Kickstarter (pending manufacturer quotes) **Distribution:** Kickstarter campaign

Hardware Specifications

Spec	Value
Display	Good Display GDEH0576T81
Size	5.76" diagonal
Resolution	920 x 680 pixels
DPI	198
Color depth	B&W with 2-bit grayscale (4 levels)
Driver IC	SSD2677
Active area	117.7 x 87.0 mm
Module size	125.4 x 99.5 x 0.9 mm
Refresh time	0.75s full / 0.3s partial
MCU	ESP32-C3 (Wi-Fi 4 + BLE 5)
Battery	~100 mAh LiPo
Solar	~70x100 mm amorphous-Si + AEM10941 harvester
Enclosure	~14 x 11 cm, white injection-molded plastic

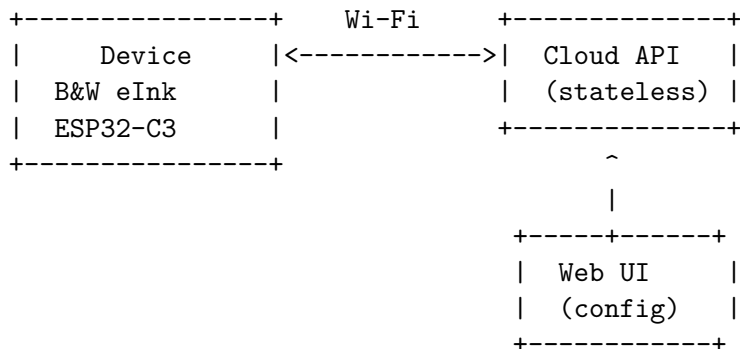
Spec	Value
Tilt detection	Two ball-in-tube tilt switches, 4 orientations
Connectivity	Wi-Fi direct, captive portal setup

Form Factor

<pre> +-----+ B&W eInk panel 920 x 680 198 DPI +-----+ </pre>	<pre> Front: display nearly fills the face minimal bezel (<=3mm) </pre>
<pre> ^ ~12-15 deg tilt body-as-stand </pre>	<pre> Back-upper: solar panel (~70x100mm) Back-lower: ESP32-C3 + battery + tilt switches </pre>

- Monolithic white housing, no hinge, no kickstand
- Body tilts ~12-15 deg so display angles toward standing viewer
- Designed to survive a 4-foot drop (bumper layer, radiused corners, recessed display)
- Pinhole reset on back for factory reset (press 5s with paperclip)

Software Architecture



Pure function API: Config JSON in, framebuffer binary out. No server-side state, no user accounts.

Server-side rendering: Device sends config to cloud API, gets framebuffer back. Device caches last framebuffer in flash for offline resilience.

Config delivery:

1. First-time: captive portal (device broadcasts **InfoBento-XXXX** SSID, user uploads config JSON + Wi-Fi credentials)
2. Ongoing: device polls **infobento.com/api/config/{device-id}** for config updates
3. Config stored in ESP32 NVS; framebuffer cached in flash

Rendering Pipeline

Stage	Detail
Input	BentoConfig (Zod-validated JSON)
Layout	Vertical stack + horizontal splits, configurable padding (0-10), content-aware height allocation
Font	Inter TTF via opentype.js, configurable 8-42px body, hero = body x 2.6
Borders	SDF-antialiased rounded rectangles, configurable corner radius (0-10)
Contrast	3-tier: hero text (dark grey), important body (black), metadata (light grey)
Output	2-bit grayscale framebuffer (156,400 bytes for 920x680)
Export	PNG with 4-level grayscale mapping

Box Types (17 total)

Category	Types	Data Source
Weather	Current weather, 8-hour forecast, 8-day forecast	Open-Meteo (free, no key)
Environment	Sunrise/sunset, air quality, moon phase	Open-Meteo + local computation
Personal	Countdown, year progress, date	Local computation
Content	Quote, text	ZenQuotes (free) / user input
Utility	QR code, tasks, calendar, habits	User input
Data	Stocks, world clock	TBD / local computation

All box types work without accounts, API keys, or subscriptions.

Web Editor

- Live preview (server-rendered PNG, landscape + portrait)
- Debounced auto-fetch for location-based boxes (500ms)
- Browser geolocation on first load (auto-detect user's city)
- Auto-fetch random quote on box add
- Configurable: font size (8-42px), corner radius (0-10), display padding (0-10)
- Show/hide box headers toggle
- Config persists in browser localStorage
- JSON import/export for portability

- No account required
- Deployed at infobento.com

Power Budget

Component	Active	Sleep	Duration
B&W eInk full refresh	~25 mA	0 mA	~0.75s
Wi-Fi connect + fetch frame	~70 mA	~10 uA	~10-20s
MCU active (during refresh)	~5 mA	~5-10 uA	~15s

Per refresh cycle: ~15-20s active, ~0.8-1.2 mAh **Daily budget (2 refreshes):** ~2-3 mAh **Solar harvest:** 5-15 mAh/day (moderate indoor light through window) **Margin:** 3-12 mAh/day surplus

User Classes

Desk Decorator (primary): Values aesthetic, minimal desk accessories. Wants a bento-box-sized display showing a quote, countdown, or QR code — standing on its own, solar-powered.

Daily Glancer (secondary): Busy professional who wants ambient weather, schedule, and quotes visible without picking up a phone.

Gift Giver: Wants a thoughtful, personalized tech gift that works immediately. Pre-configures the device via web UI, includes a card with setup instructions.

Competitive Positioning

	Tidbyt	InfoBento
Display	64x32 LED matrix	920x680 B&W eInk, 198 DPI
Feel	Glowing, animated	Paper-like, still
Power	Wall outlet required	Solar-powered, no cable
Setup	App + account	Web page, no account
Price	\$179	\$30-40 target

Tagline vs Tidbyt: “Tidbyt glows. InfoBento reads.”

Key Design Principles

1. **Zero Device Interaction** — No buttons, no app, no charging ritual. Configure once, glance forever.
2. **Pure Function Architecture** — Stateless API. Same config in, same frame out. No server-side state.
3. **Free by Default** — All 17 box types work without accounts or API keys.

4. **Grayscale Elegance** — Four shades used with intention. Hero data, body text, and metadata each earn their contrast level.
 5. **Web-Only Configuration** — Browser is the only config surface. No native app for v1.
 6. **Solar-Powered Counter Display** — Monolithic body, no hinge, no kickstand. Body is the stand.
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Open Items

- **Hardware validation:** Dev kit (ESP32-L + GDEH0576T81) on order. Validates grey rendering, refresh speed, viewing angle.
 - **Grey fallback:** If 2-bit grey looks bad on hardware, Floyd-Steinberg dithering to 1-bit as fallback (#56).
 - **Price validation:** \$30-40 target needs manufacturer quotes at Kickstarter volume.
 - **Firmware:** Captive portal provisioning (#39), Wi-Fi connect, config poll, framebuffer write, deep sleep cycle.
 - **Enclosure:** SCAD model (#50) for ~14x11cm housing, <=3mm bezel, tilt switch cavities, pinhole reset.
 - **Content-aware layout:** Height allocation for all 17 box types, not just quotes (#54).
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Monorepo Structure

infobento.com/

packages/core/	Types, layout engine, validation (Zod)
packages/renderer/	2-bit grayscale framebuffer generation
packages/api/	Hono server (stateless API + static files)
packages/web/	Vite web editor (vanilla JS, no framework)

Current version: v0.9.0 **Tests:** 145 passing across 20 test files **Quality gate:** `npm run build && npm test && npm run lint && npm run format:check`

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